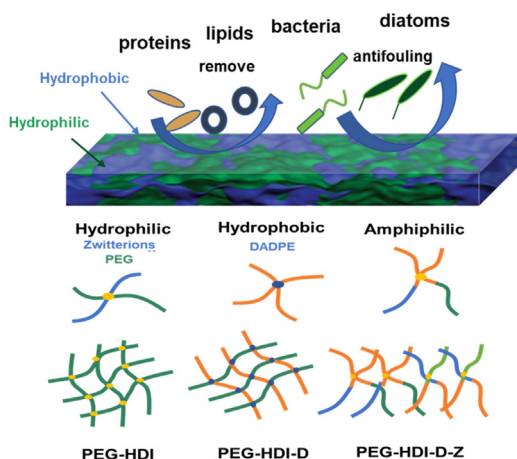


Regular Article

Marine antifouling coatings with surface topographies triggered by phase segregation

Xin Su^a, Ming Yang^a, Dezhao Hao^b, Xinglin Guo^{a,*}, Lei Jiang^b^a Key Laboratory of Science and Technology on High-tech Polymer Materials, Chinese Academy of Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, China^b Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, Beijing 100190, China

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 7 March 2021

Revised 6 April 2021

Accepted 7 April 2021

Available online 15 April 2021

Keywords:

Antifouling

Coatings

Surface structures

Phase segregation

Zwitterions

ABSTRACT

Marine biofouling is a ubiquitous and longstanding challenge that causes both economic and environmental problems. To address this, several antifouling strategies have been proposed, such as the release of biocidal compounds or surface chemical/physical design. Here we report a coating with surface structures (chemical heterogeneity) triggered by phase segregation, which endues the good antifouling properties, alongside robust mechanical properties, low underwater oil adhesion, and excellent optical transparency. This is achieved by arranging the hydrophobic and hydrophilic components to control the assembly and phase separation under the cross-linking and localized swelling process. The structure designs are based on the poly(ethylene glycols) (PEG), zwitterions, and hydrophobic components, which may lower the entropic and enthalpic driving forces for the adsorption of the marine organisms. Our approach could provide an effective way of manufacturing novel coating with amphiphilic micro/nanodomains structure, particularly for the marine industry. And we also showed that the coatings were stable under different temperatures and shear environments. To illustrate the applicability of such a robust coating in marine biofouling, we demonstrated significantly reduced algal adhesion and barnacle attachment in the sea ($p < 0.01$). We envision that this work will provide great potential for the application in antifouling marine coatings.

© 2021 Elsevier Inc. All rights reserved.

* Corresponding author.

E-mail address: xlguo@iccas.ac.cn (X. Guo).

1. Introduction

Marine biofouling, the colonization on the surface of materials by undesired marine organisms such as microorganisms, barnacles, and seaweeds, has caused serious impacts for the naval industry and marine aquaculture, with a deterioration of the surfaces, increased roughness and fuel consumption, and limited the performance of devices.[1–3] And it makes it possible that marine species would be introduced into non-native environments through ship transport.[4] Marine biofouling has been and currently is globally important due to the environmental penalties and major economic concerns.[5,6] In the last few decades, great efforts have been spent on developing efficient antifouling coatings through biocide-containing paints.[7–10] However, the extensive employ of toxic compounds raised the concerns of health and environment, and also leading to stricter regulations.[11,12] Recently, with the advancement of nanotechnology and polymer science, several novel strategies have been developed for marine anti-biofouling, such as the low-surface-energy Fouling Release Coatings (FRCs), amphiphilic, (super)hydrophilic, and topographic (structured) surfaces.[12–19] Despite decades of intense research, these surfaces are, however, still plagued with challenges that restrict their practical applications: i) the process to design and prepare a stable antifouling coating, and ii) the vast diversity of fouling organisms in the marine environment.[20–22]

In this respect, the cutting edge in creating an effective fouling-resistant coating is currently to control the interface wettability.[1,19,23] One of the most important strategies is creating hydrophilic coatings by the polyethylene glycol (PEG)-based derivatives and zwitterionic compounds through forming a thin water layer.[20,24,25] In addition, the construction of the underwater superoleophobic interface also provides a new anti-fouling vision.[26,27] Such surfaces have low adhesion to protein, lipids, and microorganisms.[28] More recently, topographic coatings emerged as an interesting option to prevent the adhesion of fouling by affecting the topographical interactions between fouling organisms and material surfaces.[29,30] However, these approaches, while promising, suffer from inherent limitations of the wettability changes (e.g., oxidation of hydroxyl group of PEG), and fabricating a stable surface morphology with a large area is still a critical challenge.[31] Furthermore, we consider that it is impossible that non-fouling methods based on coating designs incorporating a single attribute will solve the problem due to the highly turbulent marine environment.[32]

To address the aforementioned issue, here we report a novel strategy to create a polyurethane coating with amphiphilic micro/nanodomains structure triggered by phase segregation when submerged in water. These topographical structures depend highly on the process of crosslinking and localized swelling by the hydrophilic/hydrophobic component. The purpose is achieved to create surfaces with compositional (chemical) heterogeneity at the micro/nanoscale through the thermodynamically driven, phase segregation, and localized swelling of polymer assemblies. The amphiphilic micro/nanodomains on the surface may lower the entropic and enthalpic driving forces for the adsorption of the marine protein and glycoprotein bio-adhesives.[33,34] We show that the polyurethane coatings are insensitive in a complex actual environment. Both the settlement of alga in the laboratory and barnacles in the actual marine demonstrates the excellent anti-fouling performance of the coatings. Our approach could provide an effective and simple way to form PU coatings with complex topographies and different chemical composition domains in both nanoscopic and microscopic dimensions for marine anti-biofouling.

2. Experimental Section

2.1. Materials

Hexamethylene diisocyanate (HDI, AR), poly(ethylene glycol) (PEG, Mn = 1000), AR, 1,3-propane sultone and N-Methyldiethanolamine (MDEA, AR) and stannous isooctanoate were purchased from Sigma-Aldrich. (Shanghai, China). Perylene red and 4',6-diamidino-2-phenylindole (DAPI) were obtained from TCI (Shanghai, China). Triethanolamine (TEOA, AR) and 4,4'-Diaminodiphenyl ether (DADPE, AR) were purchased from Aldrich. N,N-Dimethylformamide was distilled from CaH₂ prior to use. The water used throughout all experiments was purified through a Millipore system. All reagents were used as received.

2.2. Preparation the coating of PEG-HDI

HDI (3.36 g, 0.02 mol), PEG (10 g, 0.01 mol) were separately dissolved by 6 mL of N,N-Dimethylformamide (DMF) in a dry nitrogen atmosphere. Next, the PEG solution, Triethanolamine (0.149 g, 1 mmol) and stannous isooctanoate (0.4 g, 1 mmol) were slowly added dropwise over a period of 30 min to the HDI solution. The reaction mixture was heated at 60 °C for 0.5 h. Finally, the prepolymer solution was diluted with 20 mL DMF and sprayed on the substrate (PVC, glass), and the cross-linking and curing reaction occurs fully at 80 °C.

2.3. Preparation the coating of PEG-HDI-D

The method of synthesized precursor of prepolymer solution was same as PEG-HDI. Additionally, 4,4'-Diaminodiphenyl ether (DADPE) (2 g, 0.01 mol) was dissolved with 4 mL DMF and added dropwise the system. After 0.5 h, pre-polymerization was obtained at 60 °C in a dry nitrogen atmosphere. Then spray it on the substrate and cured for one hour at 80 °C.

2.4. Preparation the coating of PEG-HDI-D-Z

Similarly, N-Methyldiethanolamine (MDEA) (0.89 g, 0.01 mol) and 1,3-propane sultone (1.22 g, 0.01 mol) were added dropwise the system of the prepolymerization solution of PEG-HDI-D. The reaction mixture was heated at 60 °C for 0.5 h, finally, the prepolymer solution was sprayed at the substrate and cured for 1 h at 80 °C.

Typically, all the samples were regularly refreshed in the water to remove the DMF and the unreacted compounds.

2.5. Instruments and characterizations

The FTIR measurements were carried out on a TENSOR-27 ATR-FTIR spectrometer in the scanning range from 4000 to 400 cm⁻¹ for 32 scans at a spectral resolution of 4 cm⁻¹ in the transmission mode. The ¹H NMR spectra of zwitterions were recorded on Bruker AV400 spectrometer using D₂O as the solvent. The chemical composition of the samples was obtained by X-ray photoelectron spectroscopy (XPS, ESCALAB250XI, USA). All spectra were calibrated by setting the signal of the aliphatic C signal at 284.8 eV. The chemical functional group and microscopic surface topography were characterized by the Nano In Situ Infrared Microscope and optical surface profiler (NT9100M, Bruker). The sample photographs were obtained using a digital SLR camera (Canon EOS70D).

The mechanical properties were measured using an electronic universal testing machine (UTM4103, Shenzhen SUNS Technology Stock Co. Ltd., China).

Shear-stress characterization using a rheometer: Generally, all the samples were prepared in a uniform shape and fitted into a self-designed holder on the rheometer platform (MCR 302, Anton Paar).

Friction and wear test: Bruker Nano Surfaces (UMT-TriboLab), the coating was tested by a pressure sensor and the set force is 10 mN. Reciprocating once every four seconds.

Water contact angle (CA) in the air and oil CA underwater were measured by the optical contact angle measuring system (OCA40 Micro, Germany). Dataphysics Instruments GmbH, Germany) at ambient temperature by placing about 2 μ L of water droplets onto the sample surface in the air. The oil adhesion is underwater obtained through the dynamic contact angle meter DCAT-21, oil of 1,2-dichloroethane was dropped onto the hole bent by the copper wire underwater, and then relevant oil adhesion force was measured. The average value from at least five measurements performed at different positions on the same sample was used as the contact angle. The hydrophilic /hydrophobic microdomain was observed using CLSM (Olympus FV1000-IX81, Japan). The hydrophilic parts were marked with DAPI and the hydrophobic parts were dyed with Perylene red. Dye was introduced during the synthesis of polyurethane coating.

2.6. Algae attachment assay and long-term marine anti biofouling test

The antifouling properties of the coating were characterized by the diatoms *Navicula. sp.* and *Nitzschia closterium*. Diatoms were purchased from Xiamen Pulan Biological Technology Co., Ltd. The main components of diatom suspension are nitrogen, phosphorus, vitamins and trace metals. Adhered algae were analyzed by auto-fluorescence image. All samples were immersed in seawater for three days to reach equilibrium before starting the antifouling tests. In addition, we used Dunett and Turkey data analysis methods in each group of samples without significant toxicity after immersing in seawater for one day. We also using LSD to analyze samples, there was obvious toxicity. However, the toxicity was significantly reduced when increasing the equilibrium time in seawater to three days. Typically, samples were placed in a six-pore plate, and 10 mL diatom suspension was added to each well. After 24 h (12 h light and 12 h dark), at a temperature is 23 °C, the samples are washed with seawater to remove unattached diatoms. In the fouling release experiment, the hydrodynamic force is simulated by a syringe pump (TS-2A/L01107-2A). All samples were scoured on a water flow with a rate of 2 m/s for 30 s. Two independent experiments and two parallel samples for the settlement of *Navicula. sp.* and *N. Closterium*, and each sample took a total of 20 measurement points of the samples were evaluated. The area is about 2*2.5 cm. The samples were observed using a fluorescence detector (DP80). The number of adherent algae is counted through image J software. The long-term antifouling test in the marine environment was tested in Da Lian (China) for 4 months (from May to Sep.) Three samples for each type were evenly sprayed on the PVC substrate (10*15*0.5 cm, actual area is 9*14 cm). And the samples were immersed in seawater and balanced for one month. All samples were fixed in one iron shelf, and the depth is about 0.2–1 m, the samples are facing the sunlight.

3. Results and discussion

3.1. Fabrication of the coating with hydrophilic/hydrophobic domain

In our experiments, we have developed a novel strategy to fabricated a coating with unique topography consisting of a chemically heterogeneous structure (hydrophilic/hydrophobic domain) triggered by phase segregation. To obtain a polyurethane coating

with amphiphilic structures, three key chemical compositions were introduced, including hydrophilic PEG, zwitterions (DEMAPS), and hydrophobic 4,4'-Diaminodiphenyl ether (DADPE). As shown in Scheme 1b, first, we prepared the polymer precursor solution by using DADPE, PEG, MDEA, and HDI. Next, 1,3-propane sultone was added to the formed solution to produce polyurethane containing zwitterions in-situ reaction. Crosslinking of the coatings was done with thermal curing via the reaction between the isocyanate groups and the hydroxyl groups in both the triethanolamine precursors and the PEG chains, producing a polyurethane network. After the further reaction, we obtained the final product and then forming a coating on the different substrate surfaces through the spraying method. Finally, the coating with hydrophilic/hydrophobic domain was formed when immersed in water via localized swelling. The antifouling mechanism was shown in Scheme 1a, on the one hand, hydrophobic components of DADPE would assemble and shrink when immersed in water, while the hydrophilic parts of zwitterion and PEG have aggregated and produced a swelling phenomenon. Meanwhile, the hydrophilic part achieves the purpose of anti-fouling by forming a hydrated film. Moreover, the hydrophilic/hydrophobic micro/nanodomains may lower the entropic and enthalpic driving forces for the adsorption of the marine bacterial, algae, protein, lipids, and other organisms.

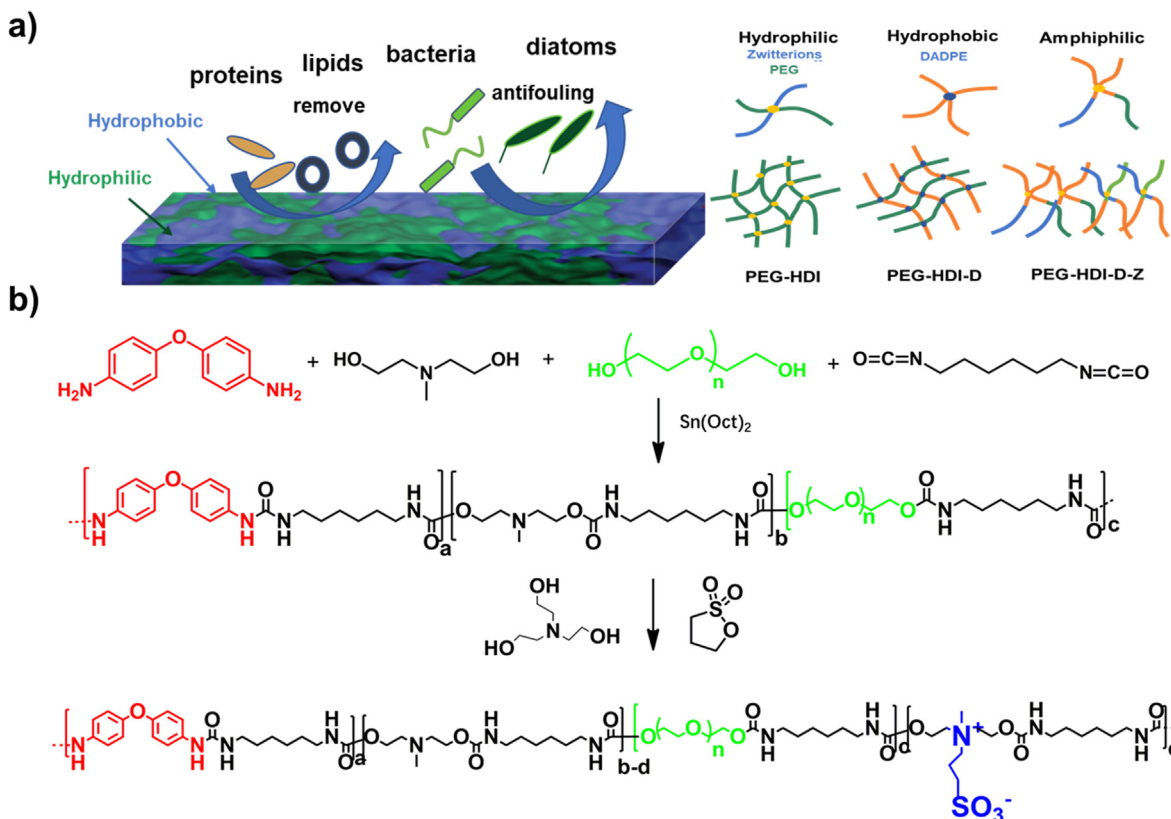
Creating surfaces with compositional (chemical) heterogeneity at the micro/nanoscale for antifouling-release purposes has been a recent trend.[35–37] We confirmed the good antifouling performance of the coating and studied the mechanism. We think that the hydrophilic/hydrophobic microdomains contained zwitterions provided an important value. The coating of the surface complexity would either have a deterrent effect on the settling stages of fouling organisms or would be unfavorable for the adsorption of adhesive proteins.

3.2. Characterization of the chemical structures

A complete characterization by FT-IR, NMR, XPS was crucial to discriminate the polymer composition and characteristics (characterization details are given in Section 2). The chemical structure of the coatings was analyzed with ATR-FTIR by measuring the obtained film after curing, the result spectrums were shown in Fig. 1a. After the curing reaction, the characteristic vibration bands of the isocyanate about 2300 cm^{-1} and hydroxyl groups about 3500 cm^{-1} were no longer present and the new vibration bands assigned to the benzene and urethane bonds formation at 850 cm^{-1} , 1500 cm^{-1} , and 3400 cm^{-1} , which also verified the existence of hydrophobic DADPE. For zwitterion of DEMAPS, the characteristic peaks of quaternary ammonium and sulfo groups are present at 1170 and 1039 cm^{-1} . Fig. S1 showed the typical ^1H NMR spectra of the zwitterion compound. We can observe the peak at d (3.1 ppm) and e (2.0 ppm) in zwitterion attributed to the ring-opening addition reaction of 1,3-propane sultone. To further investigate the incorporation of the different components in the cross-linked network, Fig. 1b and c illustrate the wide-scan XPS spectra and the typical S 2p XPS curve-fitting spectra of PEG-HDI-D-Z. A peak corresponding to S 2p appears at approximately 167 eV, which is attributed to the sulfonate groups of zwitterions, indicating an efficient introduction.

3.3. Characterization of the chemically heterogeneous structure

Amphiphilic nanostructured coatings play a critical role in antifouling.[33] Designing coatings for AF/fouling-release has been developed to create surfaces with compositional (chemical) heterogeneity at the micro-nano scale through the phase segregation of polymer assemblies and localized swelling.[38,39] In this



Scheme 1. Schematic of methods for the antifouling coating. (a) Schematic illustration of the marine antifouling mechanism. (b) Synthesis of the polyurethane coating.

work, we designed a new way to create the hydrophilic/hydrophobic domain by combining the swelling phenomenon and phase segregation. In our system, PEG and zwitterions were employed as the key constituents due to their excellent antifouling performance.[40] To verify the surface structure of the hydrophilic and hydrophobic domains, the surface morphology and the corresponding chemical composition are determined by a nano-infrared microscope (Fig. S2). The high area (blue area) represents the hydrophilic domain, and the low area (yellow area) represents the hydrophobic domain. From the infrared spectrum, we can clearly see a weak increase in the absorption strength of the hydrophobic group (benzene ring in DADPE) 1550 cm^{-1} for the blue line and while the green line indicates the group of amide for the hydrophilic domain. In addition, the surface topography is also characterized by the Profil3D white light interferometer. Fig. 2a showed coatings of PEG-HDI-D-Z before swelling underwater. In fact, all coatings prepared without swelling showed a smooth and flat morphology. Fig. 2b-d showed the topography of the PU coatings with the mountain-like structure after immersing in water for three days. Typically, PEG-HDI (Fig. 2b) and PEG-HDI-D (Fig. 2c) showed a vertical height of 666 nm and 537 nm, respectively, while PEG-HDI-D-Z (Fig. 2d) showed a height of 987 nm. The area of hilly morphology became compact and orderly for the PEG-HDI-D-Z coating, compared to the scattered distribution for the PEG-HDI and PEG-HDI-D. The structure produced a clustering effect due to the swelling of hydrophilic parts and the assembly of hydrophobic interaction. We consider that the reason is that the zwitterions reduced the phase separation scale while hydrophobic component of DADPE hindered the swelling of the system, and finally formed a more well-distributed structure of hydrophilic and hydrophobic microdomains.

3.4. Surface wettability

To evaluate the hydrophilic/hydrophobic domain and further study the wettability mechanism, we performed additional measurements to provide experimental evidence regarding the mechanism of chemical hetero by the interface wettability. Moreover, the surface wettability of materials plays a vital role in marine antifouling.[41,42] As shown in Fig. 3a, the water contact angle in the air was increased with the introduction of hydrophobic components, while the oil contact angle underwater was reduced. Furthermore, the coatings become more hydrophilic with the introduction of zwitterions. The water contact angle in the air was reduced from 55° to 35° and the oil contact angle underwater was increased from 63° to 93° . It was generally known that the introduction of surface roughness can significantly enhance surface wettability. We attribute it to the self-assembly and the surface structures of the hydrophilic/hydrophobic components when immersed in water. We consider that increasing roughness would make the hydrophilic in the air and underwater oleophobic properties more obvious. In addition, Fig. 3b showed the underwater oil adhesion force-distance curves measured by using the high-sensitivity microelectromechanical balance system. And the force drop had no obvious abrupt during the process of advancing-receding cycle. The coating of PEG-HDI-D-Z showed low oil adhesion underwater. In addition, the structure of the hydrophilic and hydrophobic microdomain was characterized by using confocal laser scanning microscopy (CLSM) (Fig. 3c-e). The results also indicate the chemically heterogeneous structure after the introduction of hydrophobic components and zwitterions. The results indicated that hydrophobic components and zwitterions played a vital role in wettability both in air and water.

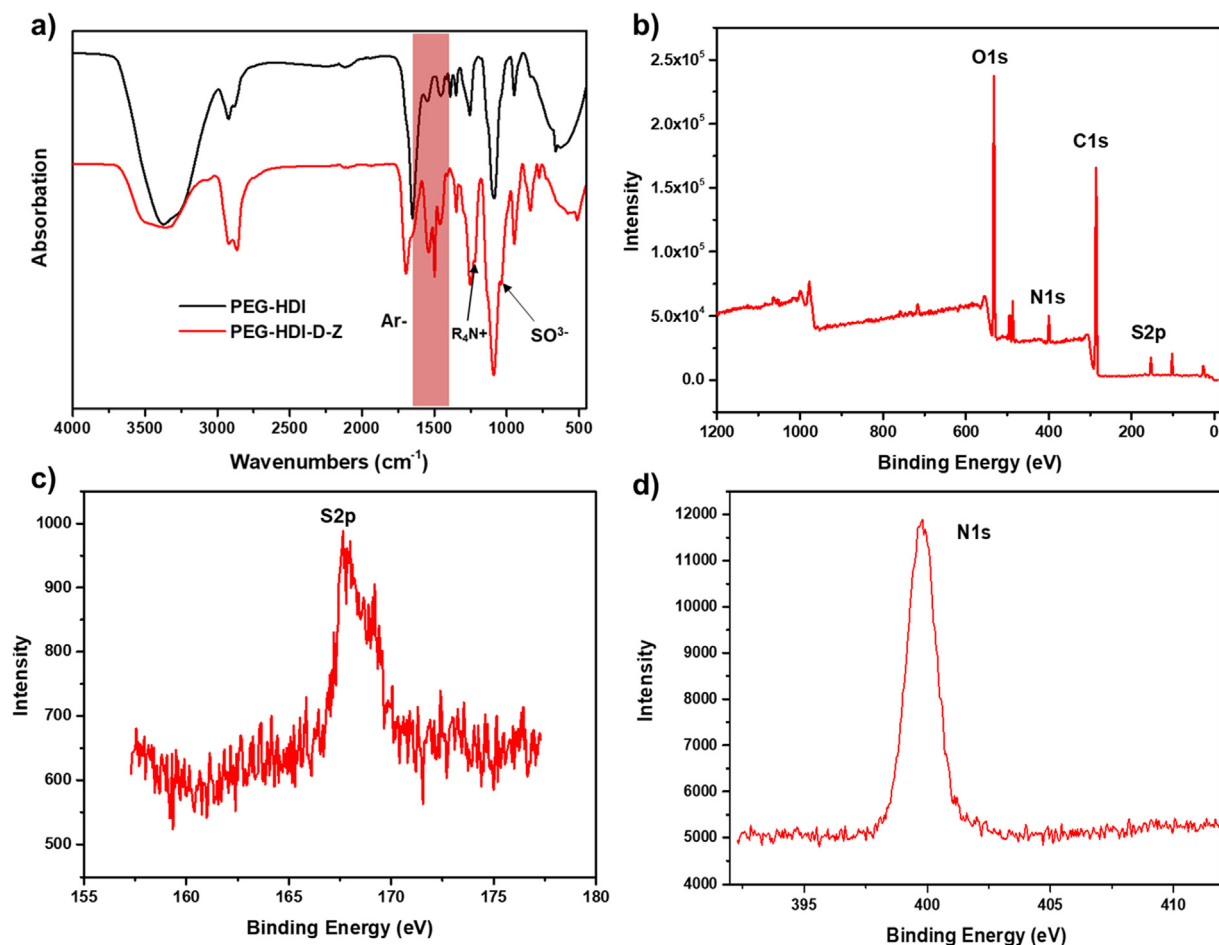


Fig. 1. Characterization of the chemical structures. (a) Fourier transform infrared spectra of coatings; (b) (c), and (d) the XPS wide-scan spectrum and the corresponding core-level high-resolution S 2p and N 1s XPS spectra of PEG-HDI-D-Z coating.

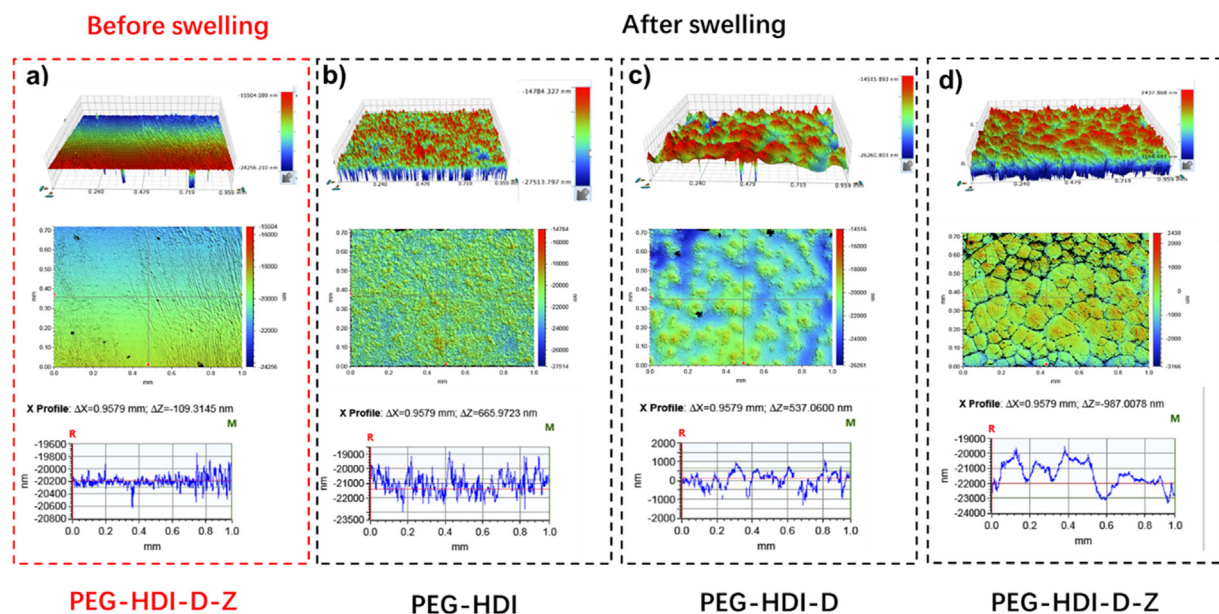


Fig. 2. Surface topography of coating before and after swelling underwater. (a) Profil3D white light interferometer images showed the uniform, dense and smooth coating before swelling. (b), (c) and (d) The images showed the morphological change after swelling underwater.

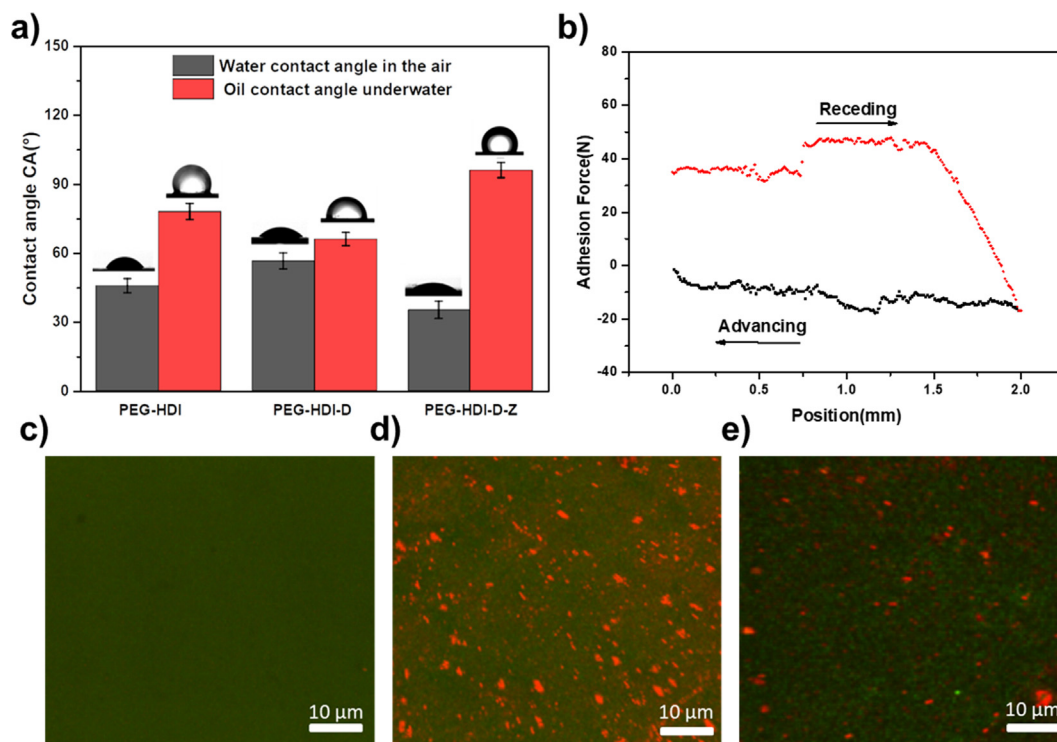


Fig. 3. Interface wettability. (a) Water contact angle in the air and oil contact angle underwater. (b) The underwater oil adhesion force–distance curve of PEG-HDI-D-Z. (c–e) CLSM images showing the structures of hydrophilic and hydrophobic micro/nano regions, the green-stained parts represent the hydrophilic regions and the red-stained parts represent the hydrophobic regions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.5. Mechanical properties

To study the mechanical properties that guarantee the extended life-time of stability in a complex environment, abrasion performance, and the long-term use of the coatings, we evaluated the performance of viscoelasticity, compression resistance, and wear-resistance. To verify the stability of the coating, the storage modulus (G') and loss modulus (G'') and were measured by Rheometer at different temperatures and frequencies. In view of the continuous hydrophilic/hydrophobic component embedded in the coating, we conjectured that such opposite character could play additional roles on the material's elastic properties. The effects of temperature and frequency on the coating stability were shown in Fig. 4a. Importantly, the PEG-HDI-D-Z showed stable elasticity over a wide temperature and shear frequency range. Fig. S1 also gives the storage modulus (G') and loss modulus (G'') of different coatings in different temperatures and frequencies. To further study the robust property, PEG-HDI-D-Z elastomer was tested under the compression-decompression for 100 cycles. As shown in Fig. 4b and 4c, the elastomer exhibited high flexibility and fatigue resistance. It was subjected to 100 loading/unloading cycles at 60% deformation amplitude and a loading rate of $1 \text{ mm} \cdot \text{min}^{-1}$. The wear resistance of the coating is extremely important in practical applications. To further study the service life as a coating for the long term, we evaluated the abrasion resistance performance with friction by multifunctional friction and wear tester. As shown in Fig. 4d, the anti-friction and wear test indicated excellent durability in a complex environment. Briefly, the coating of PEG-HDI-D-Z was abraded with a small steel ball by applying a constant force and moving a specific weight back and forth on top of the coating for a specific number of 200 cycles. The friction coefficient hardly changes sharply after 200 cycles. Fig. 4e also showed a real-time compression display sample. The results revealed that PEG-HDI-D-Z rapidly recovered their original sizes almost completely after

removing the load, sustaining the stress of 2.5 MPa (recovery ratio > 96%) after 100 cycles. The robust mechanical properties were attributed to the special hetero-network in the coating. Overall, these results demonstrate that the PEG-HDI-D-Z presents opportunities to achieve high levels of durability in the coatings or other elastomers.

3.6. Adhesion strength and transparency properties

In the search for materials with long-term durability, optical transparency, stability and the adjustable surface wettability of the coating along with excellent adhesion strength are important parameters. The adhesion strength of the coatings was estimated by measuring peeling strength after immersion in water for one week. As shown in Fig. 5a, the PEG-HDI-D-Z coating was constructed on the different substrates through a sequential blade casting method. We have selected different substrates (PVC, epoxy, primer, glass) to adapt to applications in different environments. For the PVC, epoxy, primer substrates, the peeling strength exceeded 15 N/cm . As for glass, we considered the reason for relatively low adhesion strength was the improvement of interface hydrophilicity. The increase in water absorption reduced the bonding strength of the interface. On the other hand, hydrophobic interaction would limit the swelling of hydrophilic polymer segments. In this system, to control the chemical crosslinking of the hydrophilic and hydrophobic network, a mechanically stable and water-resistant coating was established. In addition, from Fig. 5b we can clearly see the bonding strength, a 25 kg bucket could be easily hung by the PVC with the coating (area $\sim 1 \text{ cm}^2$). Moreover, optical transparency is of great significance in coatings. Fig. 5c also showed the excellent transparency of the coating synthesized by the sequential blade casting method. These results indicated the potential applications in the marine industry.

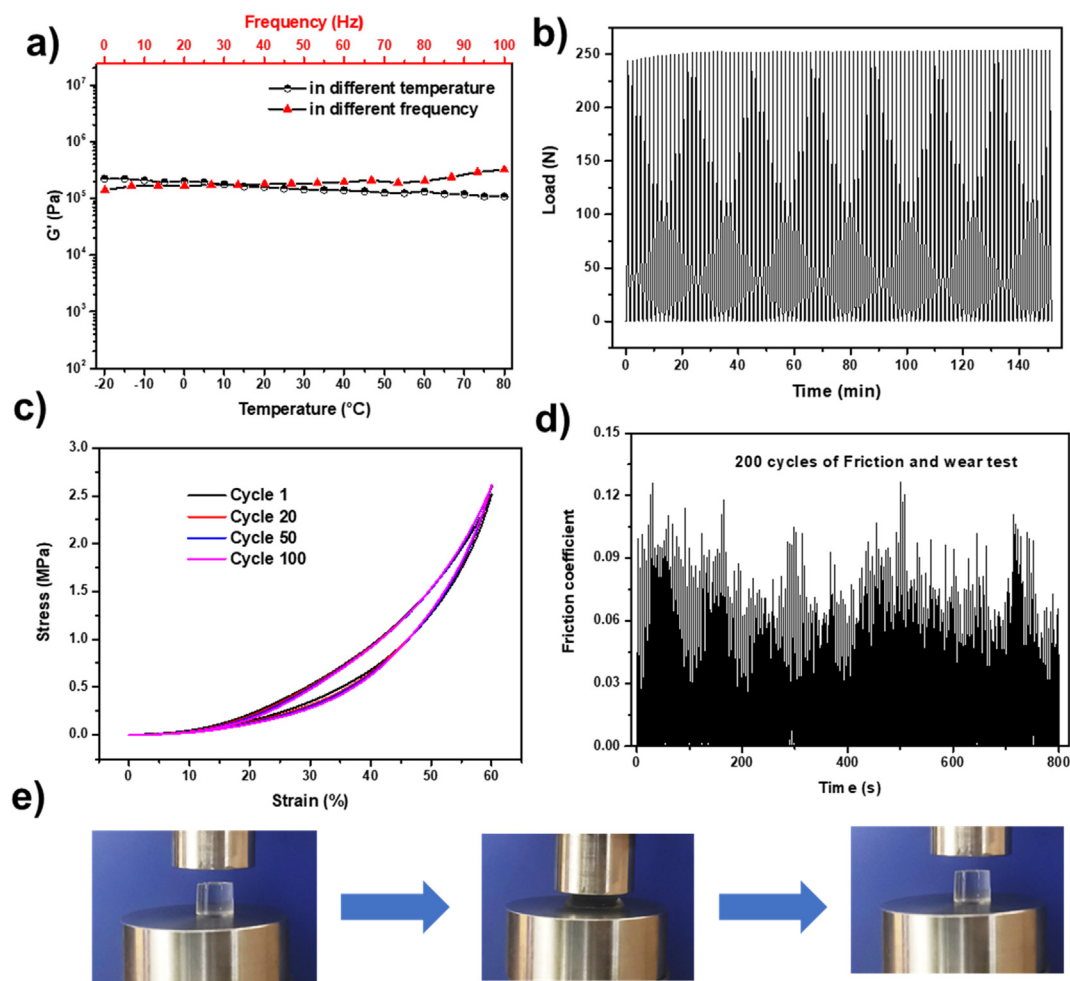


Fig. 4. Mechanical stability in a complex environment. (a) Elastic stability over a wide temperature and frequency range, the coating of PEG-HDI-D-Z on a temperature sweep in the range of -20 °C to 80 °C and frequency (Hz) from 0.1 to 100 Hz at a constant shear strain (γ) of 0.2% (b) and (c) Stress–strain curves of the PEG-HDI-D-Z at 60% strain and 100 cycles ($\varepsilon = 60\%$). (d) Friction and wear test, perform a reciprocating circular motion every 4 s, the set force is 10 mN. (e) Images showing strong mechanical stability.

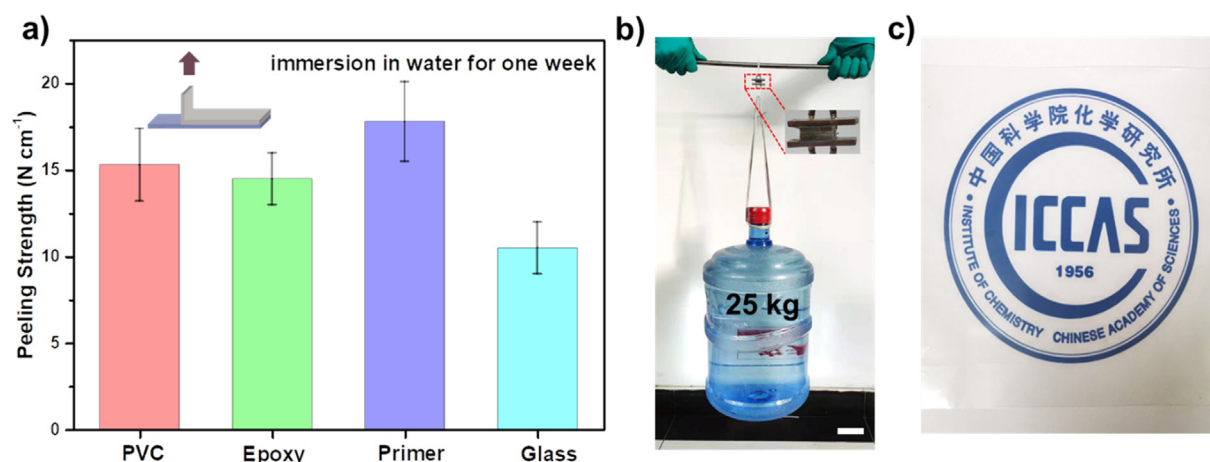


Fig. 5. Adhesion strength and transparency performance. (a) Peeling strength of the PEG-HDI-D-Z coating at the different substrates. (b) Images showing the adhesion strength at the PVC substrate, the area of the sample is 1 cm^2 . Scale bar: 5 cm . (c) The photograph showing excellent optical transparency.

3.7. Evaluation of antifouling performance

Biofouling is a complex process that involves several stages and the key factor in the design of antifouling materials is how to restrain the accumulation of physically adsorbed organic mole-

cules.[9] In this study, we design a coating to create an AF/fouling-release surface with compositional (chemical) heterogeneity at the micro/nanoscale via the thermodynamically driven, phase segregation, and localized swelling of polymer assemblies. The strategy is to combine the hydrophilic/hydrophobic compo-

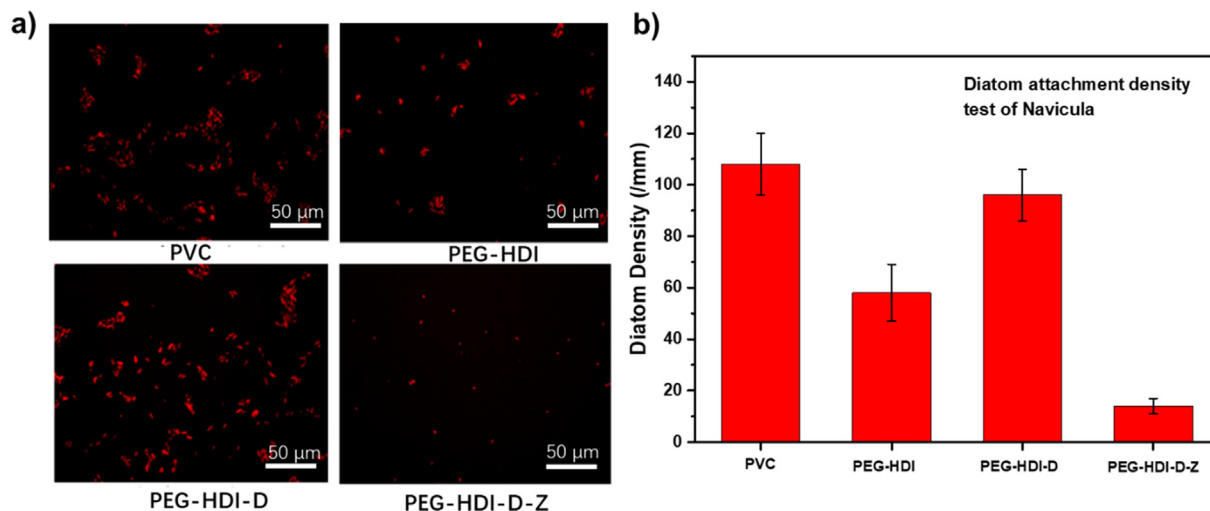


Fig. 6. Anti-diatom adhesion performance. (a) Fluorescence microscopy pictures of the densities of adhered *Navicula* sp. The data were obtained from a suspension of 10^6 cells mL^{-1} . (b) A statistical chart showing the diatoms adhesion ($p < 0.01$).

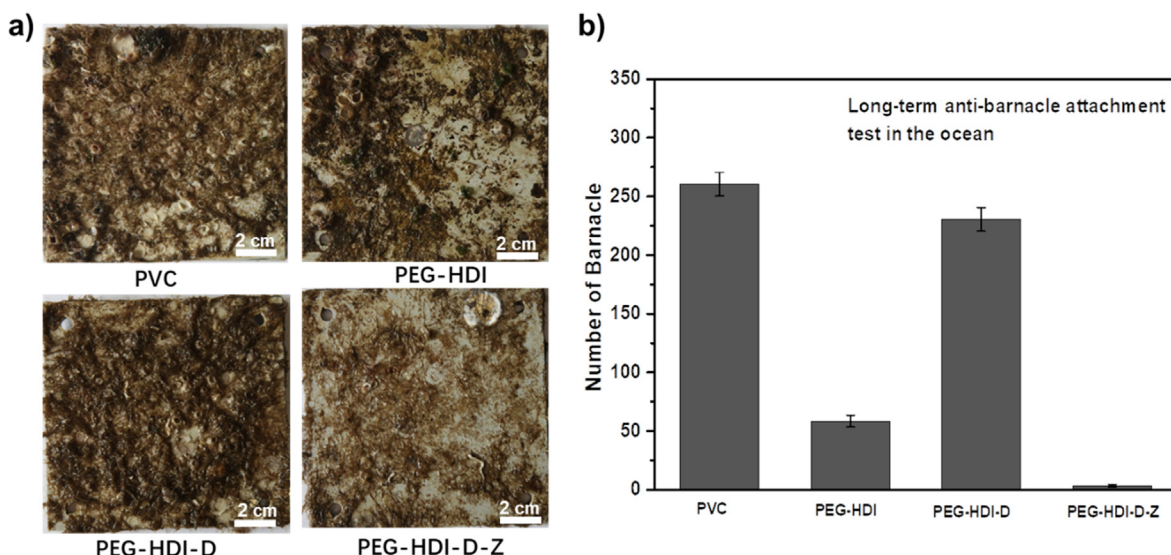


Fig. 7. Long-term actual marine antifouling experiment. (a) Photographs of the actual marine anti-biofouling performance (no cleaning). (b) A statistical chart showing the number of adhered barnacles.

ment, expressed in terms of amphiphilic domains on the surface, which may lower the entropic and enthalpic driving forces for the adsorption of the marine organisms.

A simplified laboratory-accessible way to evaluate the anti-fouling performance is to quantify the first stage of biofouling by measuring the algae adsorption on surfaces. We investigated the adhered densities of diatom by fluorescence microscopy. As shown in Fig. 6a and b, the diatom of *Navicula* sp. heavily fouled the PVC sheets as the control group, the coating showed poor performance while introducing the hydrophobic component compared to the PEG-HDI. However, the PEG-HDI-D-Z coating showed good anti-diatom performance when introducing the zwitterions. Fig. S4 also showed the excellent antifouling property for the diatom of *Nitzschia Closterium*. The result indicates the key role of zwitterions in the antifouling coating of PEG-HDI-D-Z.

In this study, we also investigated the antifouling performance of a coating in the actual marine environment. In our previous study, we select the barnacles as a typical survey object for statis-

tics in the long-term antifouling test. (see details in the Experimental Section). The results are shown in Fig. 7a and 7b. Numerous barnacles were absorbed and settled on the PVC sheets as the control group. The coating of PEG-HDI-D-Z showed the lowest barnacle attachment. It is worth noting that sludge and soft fouling can be removed by gently rinsing in seawater. The results show that the coating exhibits low diatom adsorption and good barnacle adhesion resistance, so it can be considered to have a high antifouling potential value. In addition, we also evaluated the fouling release characteristics of the coating under water flow conditions. All the coatings showed good anti-diatom property compared to the control samples of PVC (Fig. S5). Which also indicated the weak attachment of diatom.

Moreover, we think the mechanism of the inhibitory effect of such surfaces is related to entropic, topographies, and chemical properties. The amphiphilic coatings combined hydrophobic/hydrophilic components would reduce polar and hydrogen-bonding interactions with the fouling, which may prevent the

adsorption and folding of fouling organisms. Increasing hydrophobicity or reducing the underwater oil adhesion may reduce antifouling performance, while hydrophilic/hydrophobic chemical heterostructures also play a critical role in antifouling coatings. In this paper, we envision that these results could demonstrate the importance of topographies and chemically heterogeneous structure for engineering antifouling surfaces.

4. Conclusions

In summary, we prepared an antifouling coating with surface topographies structure triggered by phase segregation. The surface topography is presumably caused by localized swelling of the aggregated hydrophilic blocks when immersed in the coating in water. Owing to the distinctive hydrophilic/hydrophobic micro-nanodomain, which may lower the entropic and enthalpic driving forces for the adsorption and settlement of the marine microorganisms and meanwhile, zwitterions could form a thin water layer to prevent the attachment of marine microorganisms. Moreover, the polyurethane coating with robust mechanical stability greatly improves the application range and service life. Such a coating design substantially facilitates the development of 'multifunctional coatings' to solve the marine antifouling problem. This work highlights the importance of interface design on the construction of antifouling coating based on appropriate topography combined with a suitable amphiphilic or zwitterionic surface chemistry and also provides a simple model that could potentially spark further experimental and theoretical efforts for the application in this area.

CRediT authorship contribution statement

Xin Su: Conceptualization, Data curation, Writing - original draft, Methodology. **Ming Yang:** Data curation. **Dezhao Hao:** Data curation, Formal analysis. **Xinglin Guo:** Conceptualization, Writing - review & editing. **Lei Jiang:** Funding acquisition, Project, administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research is supported by the National Natural Science Foundation of China (U2006219), National Research Fund for Fundamental Key Projects (2013CB933000), National Natural Science Foundation (21671194, 21421061), the Key Research Program of the Chinese Academy of Sciences (KJZD-EW-M01).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2021.04.031>.

References

- [1] J.A. Callow, M.E. Callow, *Nat. Commun.* 2 (2011) 244.
- [2] R.R. Choudhury, J.M. Gohil, S. Mohanty, S.K. Nayak, *J. Mater. Chem. A* 6 (2018) 313–333.
- [3] S. Dobretsov, R.M. Abed, M. Teplitski, *Biofouling* 29 (2013) 423–441.
- [4] I. Fitridge, T. Dempster, J. Guenther, R. de Nys, *Biofouling* 28 (2012) 649–669.
- [5] N. Fusetani, *Nat. Prod. Rep.* 28 (2011) 400–410.
- [6] K.V. Thomas, S. Brooks, *Biofouling* 26 (2010) 73–88.
- [7] I. Banerjee, R.C. Pangule, R.S. Kane, *Adv. Mater.* 23 (2011) 690–718.
- [8] M. Lejars, A. Margailan, C. Bressy, *Chem. Rev.* 112 (2012) 4347–4390.
- [9] A.J. Scardino, H. Zhang, D.J. Cookson, R.N. Lamb, R. de Nys, *Biofouling* 25 (2009) 757–767.
- [10] C.M. Magin, J.A. Finlay, G. Clay, M.E. Callow, J.A. Callow, A.B. Brennan, *Biomacromolecules* 12 (2011) 915–922.
- [11] J.B. Kristensen, R.L. Meyer, B.S. Laursen, S. Shipovskov, F. Besenbacher, C.H. Poulsen, *Biotechnol. Adv.* 26 (2008) 471–481.
- [12] J. Genzer, K. Efimenko, *Biofouling* 22 (2006) 339–360.
- [13] J.L. Dalsin, P.B. Messersmith, *Mater. Today* 8 (2005) 38–46.
- [14] G.D. Bixler, A. Theiss, B. Bhushan, S.C. Lee, *J. Colloid Interface Sci.* 419 (2014) 114–133.
- [15] M. Nosonovsky, B. Bhushan, *Curr. Opin. Colloid Interface Sci.* 14 (2009) 270–280.
- [16] E. Almeida, T.C. Diamantino, O. de Sousa, *Prog. Organic Coat.* 59 (2007) 2–20.
- [17] Y. Zhang, H. Hu, X. Pei, Y. Liu, Q. Ye, F. Zhou, *Biomater. Sci.* 5 (2017) 2493–2500.
- [18] Q. Xie, Q. Xie, J. Pan, C. Ma, G. Zhang, *ACS Appl. Mater. Interfaces* 10 (2018) 11213–11220.
- [19] X. Zhu, S. Guo, D. Janczewski, F.J. Velandia, S.L. Teo, G.J. Vancso, *Langmuir* 30 (2014) 288–296.
- [20] S. Martin, P.S. Brown, B. Bhushan, *Adv. Colloid Interface Sci.* 241 (2017) 1–23.
- [21] B.R. Knowles, P. Wagner, S. MacLaughlin, M.J. Higgins, P.J. Molino, *ACS Appl. Mater. Interfaces* 9 (2017) 18584–18594.
- [22] J.F. Briand, I. Djeridi, D. Jamet, S. Coupe, C. Bressy, M. Molmeret, B. Le Berre, F. Rimet, A. Bouchez, Y. Blache, *Biofouling* 28 (2012) 453–463.
- [23] G.D. Bixler, B. Bhushan, *Philos. Trans. A Math. Phys. Eng. Sci.* 370 (2012) 2381–2417.
- [24] P.J. Molino, D. Yang, M. Penna, K. Miyazawa, B.R. Knowles, S. MacLaughlin, T. Fukuma, I. Yarovsky, M.J. Higgins, *ACS Nano* 12 (2018) 11610–11624.
- [25] C.-J. Huang, S.-H. Chu, L.-C. Wang, C.-H. Li, T.R. Lee, *ACS Appl. Mater. Interfaces* 7 (2015) 23776–23786.
- [26] T. Du, S. Ma, X. Pei, S. Wang, F. Zhou, *Small* 13 (2017) 1602020.
- [27] P. Zhang, L. Lin, D. Zang, X. Guo, M. Liu, *Small* 13 (2017) 1503334.
- [28] R. Zhang, L. Zhang, N. Tian, S. Ma, Y. Liu, B. Yu, X. Pei, F. Zhou, *Adv. Funct. Mater.* 4 (2017) 1601039.
- [29] J.T. Decker, J.T. Sheats, A.B. Brennan, *Langmuir* 30 (2014) 15212–15218.
- [30] C.J. Long, J.A. Finlay, M.E. Callow, J.A. Callow, A.B. Brennan, *Biofouling* 26 (2010) 941–952.
- [31] S. Lowe, N.M. O'Brien-Simpson, L.A. Connal, *Polymer Chemistry* 6 (2015) 198–212.
- [32] I.K. Konstantinou, T.A. Albanis, *Environ. Int.* 30 (2004) 235–248.
- [33] D.R. Calabrese, B. Wenning, J.A. Finlay, M.E. Callow, J.A. Callow, D. Fischer, C.K. Ober, *Polym. Adv. Technol.* 26 (2015) 829–836.
- [34] R. Nugraha, J.A. Finlay, S. Hill, T. Fyrner, W. Yandi, M.E. Callow, J.A. Callow, T. Ederth, *Biofouling* 31 (2015) 123–134.
- [35] E. Martinelli, S.D. Hill, J.A. Finlay, M.E. Callow, J.A. Callow, A. Glisenti, G. Galli, *Prog. Organic Coat.* 90 (2016) 235–242.
- [36] W. van Zoelen, H.G. Buss, N.C. Ellebracht, N.A. Lynd, D.A. Fischer, J. Finlay, S. Hill, M.E. Callow, J.A. Callow, E.J. Kramer, R.N. Zuckermann, R.A. Segalman, *ACS Macro Lett.* 3 (2014) 364–368.
- [37] Y. Cho, H.S. Sundaram, C.J. Weinman, M.Y. Paik, M.D. Dimitriou, J.A. Finlay, M.E. Callow, J.A. Callow, E.J. Kramer, C.K. Ober, *Macromolecules* 44 (2011) 4783–4792.
- [38] Z. Zhou, D.R. Calabrese, W. Taylor, J.A. Finlay, M.E. Callow, J.A. Callow, D. Fischer, E.J. Kramer, C.K. Ober, *Biofouling* 30 (2014) 589–604.
- [39] S.J. Stafslien, S. Sommer, D.C. Webster, R. Bodkhe, R. Pieper, J. Daniels, L. Vander Wal, M.C. Callow, J.A. Callow, E. Ralston, G. Swain, L. Brewer, D. Wendt, G.H. Dickinson, C.S. Lim, S.L. Teo, *Biofouling* 32 (2016) 949–968.
- [40] W. Yandi, S. Mieszkina, P. Martin-Tanchereau, M.E. Callow, J.A. Callow, L. Tyson, B. Liedberg, T. Ederth, *ACS Appl. Mater. Interfaces* 6 (2014) 11448–11458.
- [41] Q. Ye, T. Gao, F. Wan, B. Yu, X. Pei, F. Zhou, Q. Xue, *J. Mater. Chem.* 22 (2012) 13123.
- [42] E. Martinelli, E. Guazzelli, C. Bartoli, M. Gazzarri, F. Chiellini, G. Galli, M.E. Callow, J.A. Callow, J.A. Finlay, S. Hill, *J. Polym. Sci. Part A: Polym. Chem.* 53 (2015) 1213–1225.